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Automate the App Lifecycle with CumulusCI

David Reed and Marcel Pufal
Release Engineering | Salesforce.org



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Presenters



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Salesforce.org is a social impact center of Salesforce focused on partnering with the global community to tackle the world's biggest problems.

We believe that technology, when used for good, can change the world. We build powerful technology for, and with, our community of nonprofit, educational and philanthropic organizations



Release Engineering at Salesforce.org

By the numbers

- Manage development pipeline of **over 40 managed packages**.
- Shipping to **tens of thousands** of customer orgs every 2 weeks.
- Build **5** major open source tools:
 - **CumulusCI** automation framework
 - **Snowfakery** data generation tool
 - **MetaCI** continuous integration
 - **MetaDeploy** smart installer platform
 - **Metecho** to bring SFDX and Git to admins
- Run over **380,000** scratch org builds.
- Saving development teams est. **6-8 hours per scratch org**.



Agenda



The trail to production and the org problem

The Salesforce.org solution: portable automation with CumulusCI

Live demo

Completing the picture with MetaCI and MetaDeploy

Takeaways



Let's Get Cost Out of the Way



This is not a sales pitch

CumulusCI, Snowfakery, MetaCI, Metecho, and MetaDeploy are 100% free and open source.

Salesforce licenses and server infrastructure do cost money.

- You'll still pay Heroku, CircleCI, Jenkins, et al.
for infrastructure if you wish to use continuous integration

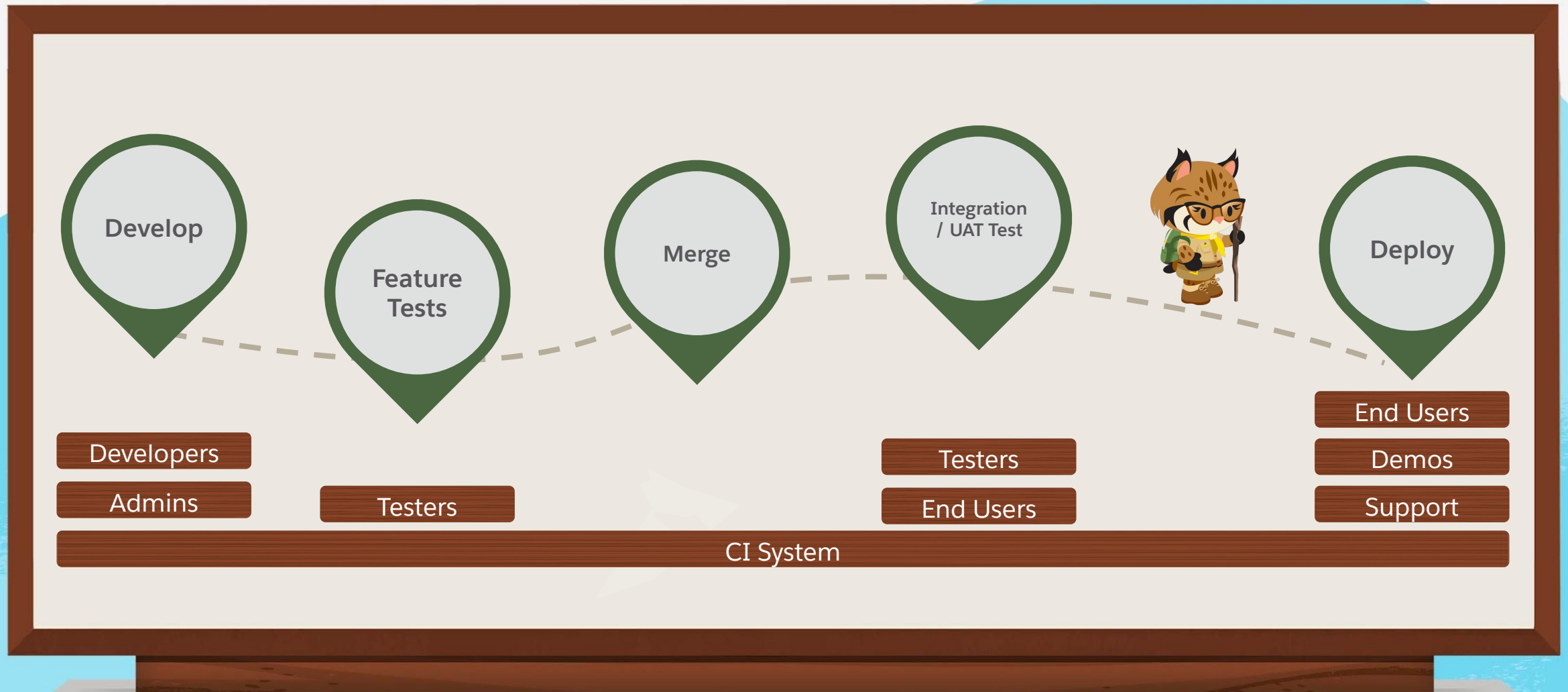
A wooden signpost with a brown, textured surface, mounted on a wooden post. The sign is set against a background of a green tree and a blue sky. The text on the sign is white and reads "Support the Salesforce open source community!".

Support the Salesforce
open source
community!

The Trail to Production

Exploring the app lifecycle

The trail to production



Where do these participants do their work?



Continuous integration must be automated.

Most CI systems do very similar steps:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.
- Execute tests (Apex unit tests, Jest tests, Robot browser tests...)
- Report results

In many organizations, these steps are so complex that they end up using a single, persistent org.

Developers, admins, testers, support, product managers, and other stakeholders usually must create environments manually.

Each user has to:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.

In many organizations, these steps are so expensive that users must make do with shared, out-of-date, or inconsistent sandboxes.

The org problem

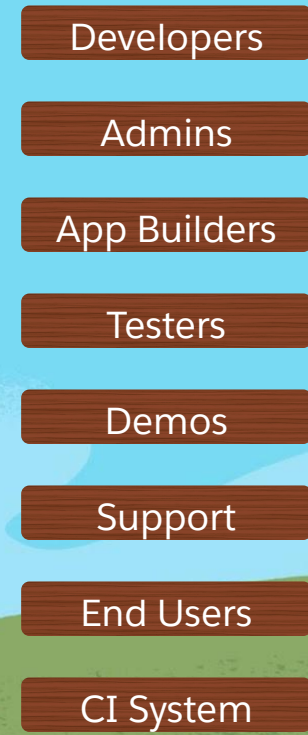
All of these roles and systems need Salesforce orgs to work in.

Building new orgs is hard.

- Traditional tools only do metadata setup.
- What about configuration? Data? Dependency management?
- What about multiple org shapes for different roles?
- CI automation is usually not “portable” (CI only).

Managing persistent orgs’ state is hard.

- Challenging to fully synchronize with source control
- Weak change management
- Users conflicting with one another

A vertical stack of eight brown rectangular buttons with white text, representing different roles and systems that need Salesforce orgs. The buttons are: Developers, Admins, App Builders, Testers, Demos, Support, End Users, and CI System. They are positioned on the right side of the slide, overlaid on a light blue wavy background that resembles a body of water, with a green hill and a tree at the bottom right.

Developers

Admins

App Builders

Testers

Demos

Support

End Users

CI System

The Salesforce.org Solution: Portable Automation with CumulusCI

Portable automation for every role,
everywhere on the trail

Basic framework



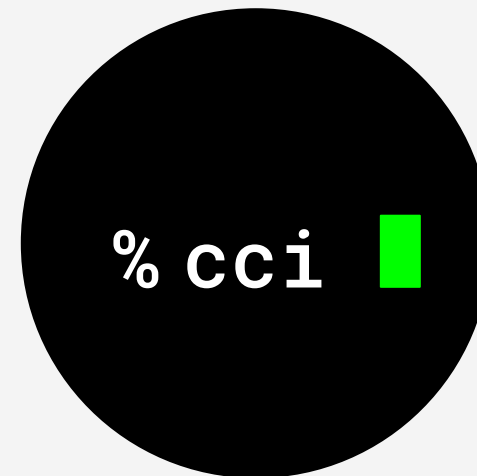
Version control as
source of truth.



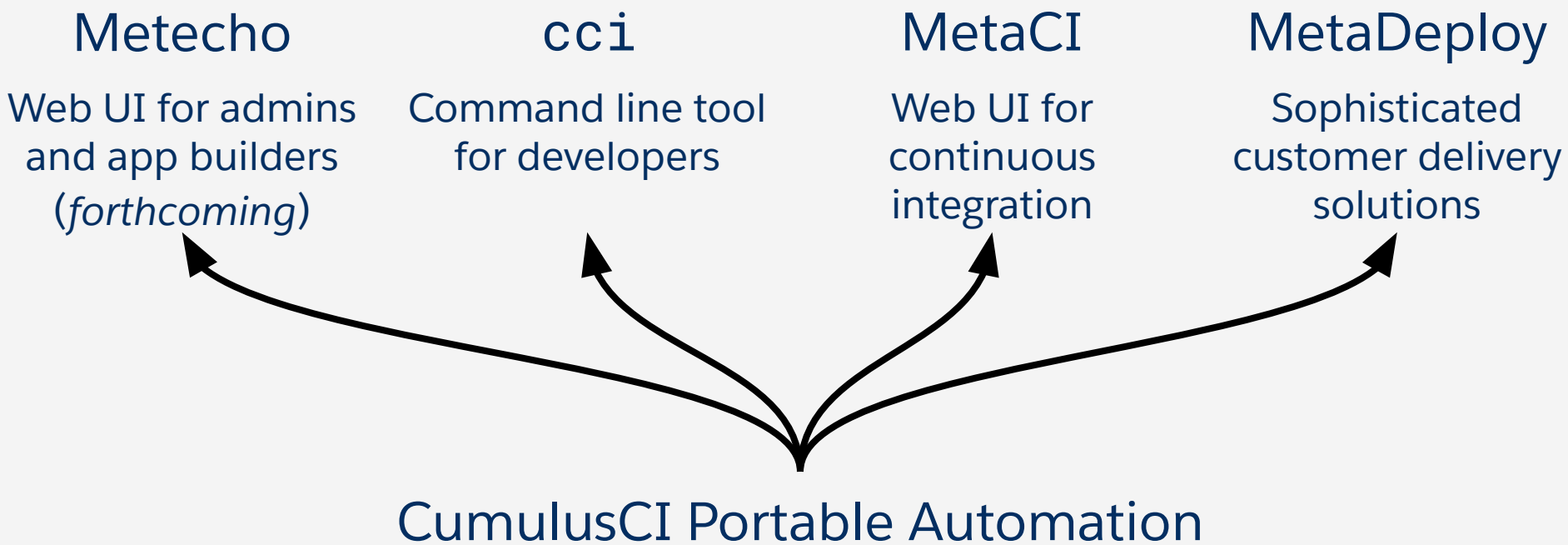
Salesforce DX scratch
orgs as location of work.



CumulusCI as the
automation engine.



Same automation, many contexts



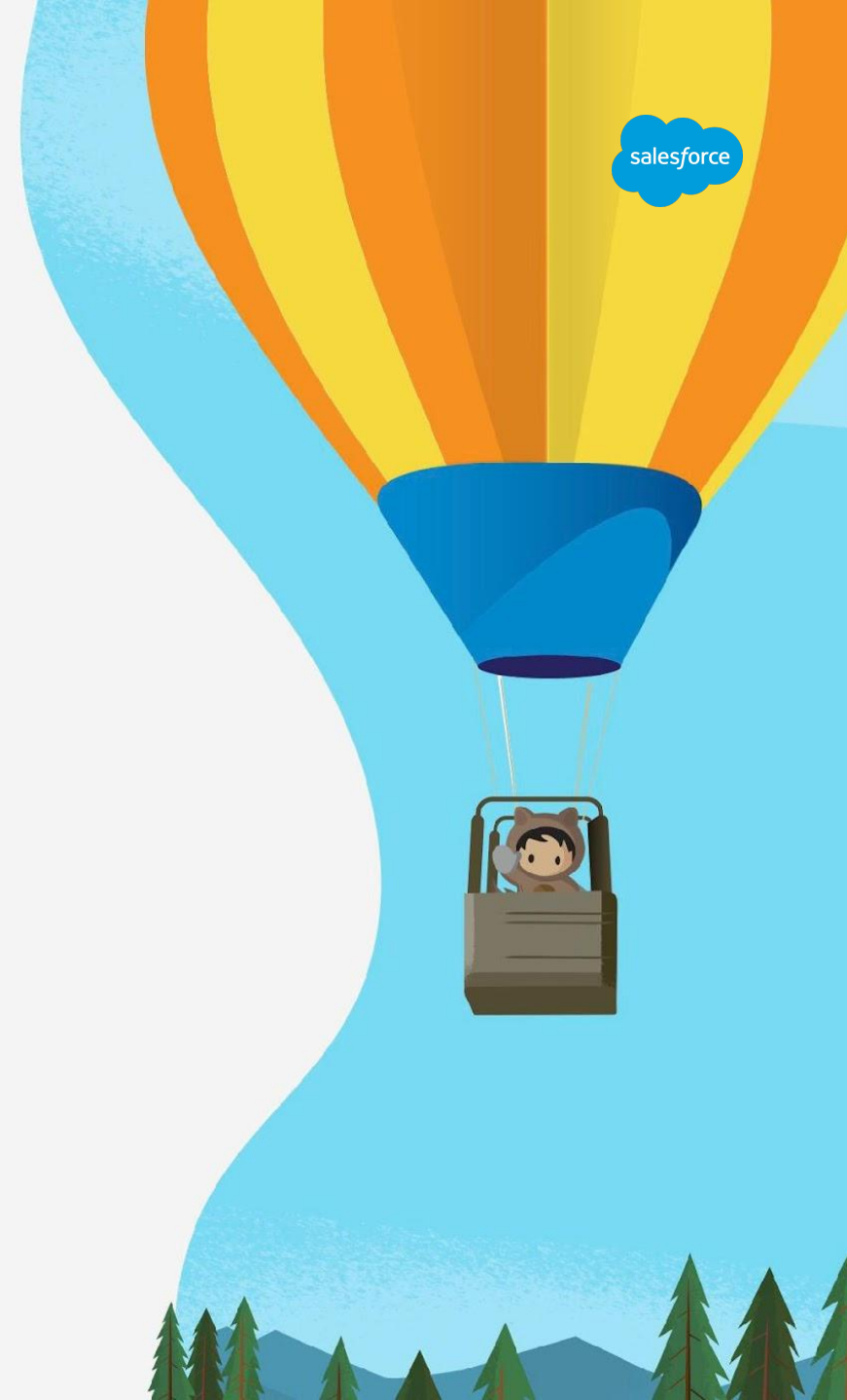
What CumulusCI does

Building orgs

Builds orgs based on repeatable recipes

- Dependency management
- Package or application installation
- Metadata deployment to tailor org
- Configuration of settings
- Data seeding
- Any task achievable with the Salesforce API

CumulusCI makes it easy to define fully-realized scratch orgs for each of your roles.



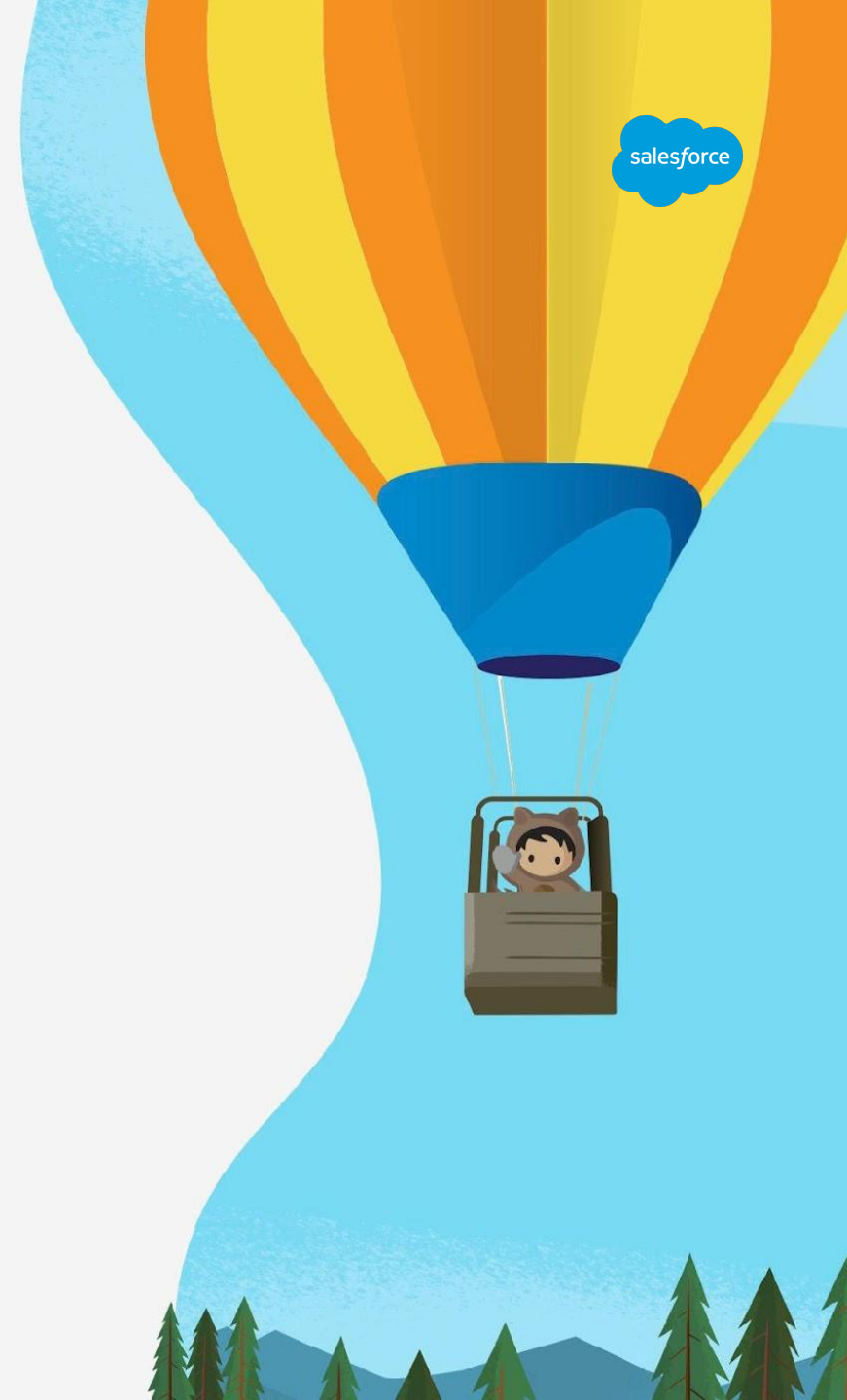
What CumulusCI does

Building apps

Provides tools for each role to use in their orgs

- Run tests (Apex, Jest, Robot browser and API automation)
- Develop and deploy code and metadata
- Load, modify, and capture data sets

CumulusCI makes it easy for developers, admins, and testers to do their work and collaborate in version control.



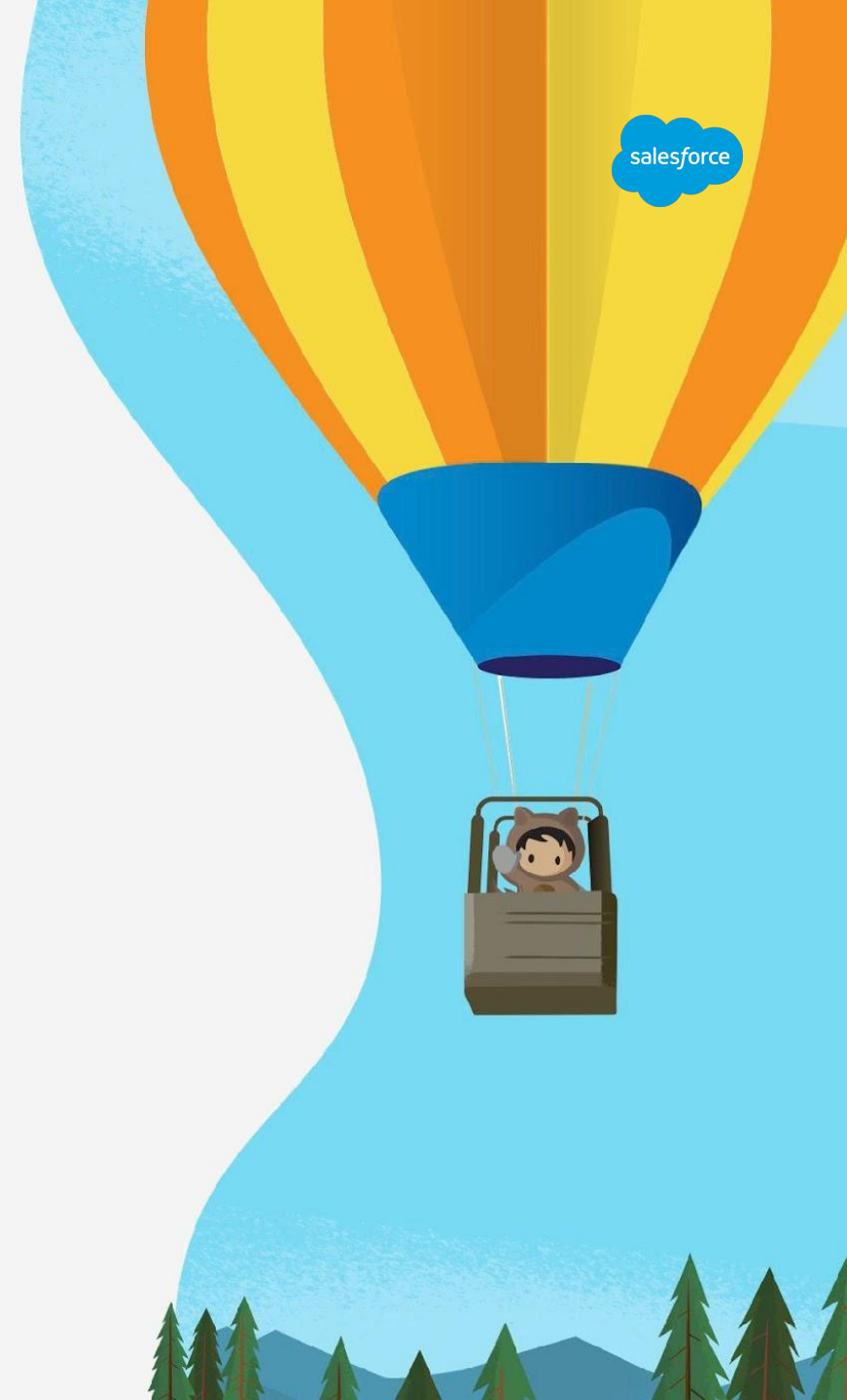
What CumulusCI does

Delivering work

Provides tools to orchestrate the entire development process

- Run continuous integration builds
 - New support for Second-Generation Packaging tests
- Execute Apex, Jest, and Robot Framework browser tests
- Automate parent-child branch merges (CumulusCI Flow)
- Upload and test beta managed packages automatically
- Upload and test managed releases

Continuous integration server recommended but not required: you can run the entire development process from your local machine.



Add Snowfakery for even more

Generate realistic test data at any volume

Generate the data you need to test and train

- Create complex relational data that looks real.
- Set your own patterns, invariants, and value distributions
- Take advantage of rich data synthesis libraries
- Load data at any volume, even tens of millions of records

CumulusCI and Snowfakery make it easy to create data-rich environments for testing and training, with no PII exposure.

```
- object: Opportunity
count:
  random_number:
    min: 3
    max: 5
fields:
  Name: ${{fake.word}} Opportunity
  CloseDate:
    date_between:
      start_date: -30d
      end_date: +180d
  Amount:
    random_number:
      min: 10
      max: 200
  StageName:
    random_choice:
      - choice:
          probability: 60%
          pick: Closed Won
```

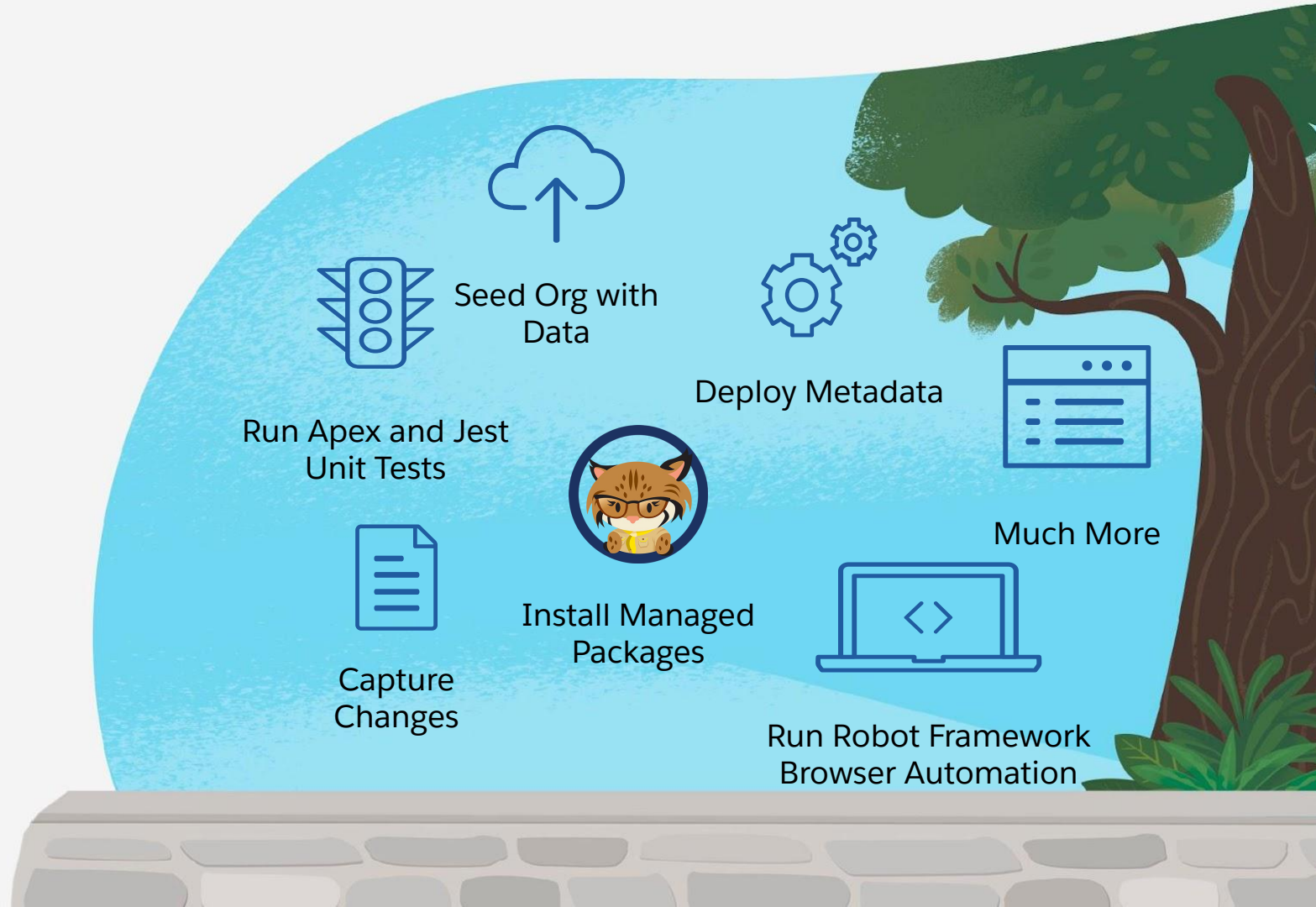
Flows and tasks

Composable building blocks for automation

CumulusCI comes with dozens of built-in tasks and predefined flows.

Tasks and flows are configurable in simple text markup.

New capabilities can be added in Python.



Customize flows and tasks in version control

`cumulusci.yml` lives in the repository alongside code

```
trailhead_org:
  steps:
    1:
      task: update_dependencies
    2:
      task: deploy
      options:
        path: unpackaged/pre/account_record_types
    3:
      task: ensure_record_type
    4:
      task: install_managed
    5:
      flow: config_trailhead_playground
```

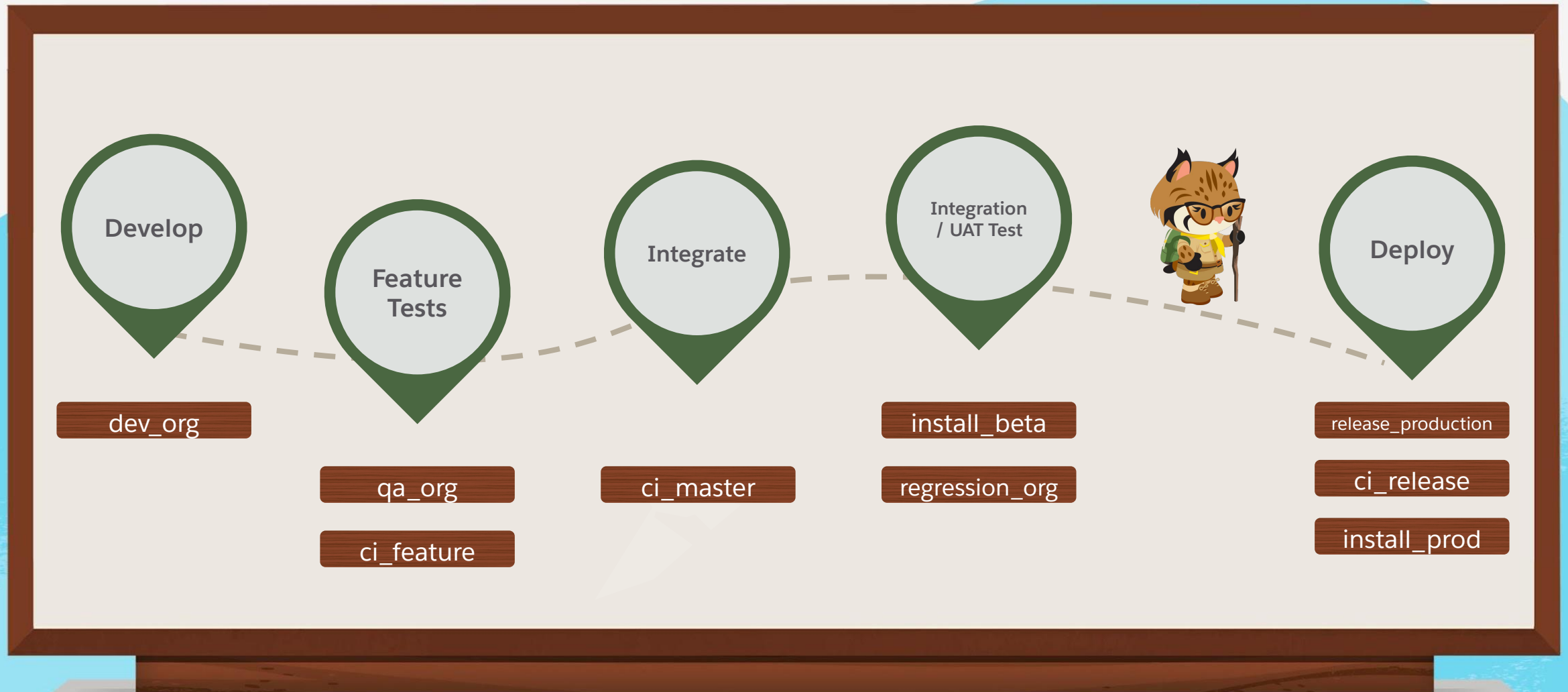


A stylized illustration of a forest scene. Two large, leafy green trees with thick brown trunks frame the top and sides of the image. In the background, there are rolling green hills and several tall, thin evergreen trees. The sky is a light blue gradient. In the bottom right corner, there are some yellow flowers and green foliage. A small bird is perched on a branch of the tree on the left.

Hands-On Demonstration

The trail to production

salesforce



Outcomes of solving the org problem



Increased velocity

- Portable automation means no org blockers or infrastructure dependencies

Increased confidence

- Rely on orgs always being configured exactly as you expect
- Test results are repeatable and reliable

Modern best practices not just baked-in, but enabled

- Source control
- Salesforce DX

Users get access to holistic environments that encourage collaboration

- CumulusCI makes it easy to capture and share work.



Completing the Picture with MetaCI and MetaDeploy

Applying portable automation to continuous
integration and customer delivery

MetaCI



Continuous integration for Salesforce projects

Orchestrate development workflow.

Easily define plans that run CumulusCI automation on specific branches and tags.

Status reported to GitHub for process insights.

Track releases and automatically generate release notes in GitHub.

 **All checks have passed** [Hide all checks](#)
3 successful checks

	 Feature - Prerelease — The build was successful Details
	 Feature Test — The build was successful Required Details
	 Feature Test - Namespaced — The build was successful Details

Latest release

rel/3.117
93f3eb9

Compare ▼

3.117

 salesforce-org-metaci released this 10 days ago

Changes



MetaCI

Support declarative users with web UI

Create scratch orgs using CumulusCI automation through a web UI.



MetaCI

Create Org

Create Scratch Org

Use this section to create a scratch org from a branch for general feedback purposes.

Create Scratch Org

Prefer to bring your own CI?

No lock-in



MetaCI makes it easy to map CumulusCI automation to your GitHub branches and tags and implement CumulusCI Flow.

CumulusCI works great in other CI systems too.

Use CumulusCI as your build tool in CircleCI, Jenkins, Travis, GitHub Actions, or whatever you prefer. Even on-premises!



MetaDeploy

Customer delivery with portable automation



Log In ▼

Select a Product to Install

Salesforce.org Products

Open Source Commons



Insights Platform Data Integrity

Version 1.14

Data Cleansing for Your Organization



Nonprofit Cloud Case Management

Version 1.20

Deliver Personalized Client Services



Volunteers for Salesforce (V4S)

Version 3.121

Volunteer management for your organization.



Nonprofit Success Pack (NPSP)

Version 3.189

The Foundation for the Connected Nonprofit



Program Management Module (PMM)

Version 1.19

Track, manage, and deliver programs



K-12 Architecture Kit

Version 1.20

A Custom Solution for K-12 Schools and Districts



Grants Management

Version 1.7

Grants Management Tools and Communities



Outbound Funds Module

Version 1.19

Track and manage outgoing funding programs.



Education Data Architecture (EDA)

Version 1.109

The Foundation for the Connected Campus

Portable Automation

Your automation drives both scratch org builds and customer delivery

A small icon representing a terminal window, consisting of three colored circles (red, yellow, green) in a horizontal row.

```
trailhead_org:
  steps:
    1:
      task: update_dependencies
    2:
      task: deploy
      options:
        path: unpackaged/pre/account_record_types
    3:
      task: ensure_record_type
    4:
      task: install_managed
    5:
      flow: config_trailhead_playground
```



Create great experiences with logic and feedback



Validate the customer org



```
checks:
```

- `when: "not tasks.check_chatter_enabled( )"`
 `action: error`
 `message: "Outbound Funds Module requires Chatter."`
- `when: "'.my.'" not in org_config.instance_url"`
 `action: error`
 `message: "Please enable My Domain in your org prior to installing."`



MetaDeploy

Smart, product-oriented installers create a great customer experience



Grants Management, 1.7

Select a Plan

Install Grants Management and Build Community - View Details

Install Grants Management - View Details

← Select a different product

MetaDeploy

Smart, product-oriented installers create a great customer experience



Steps	Type	
Install SFDO Base 1.0	 Managed Package	Required
Install Outbound Funds (Core) 1.19	 Managed Package	Required
Deploy Visual Assets	 Metadata	Required
Install Grants Management 1.7	 Managed Package	Required
Deploy Create Application Flow	 Metadata	Required
Deploy Profiles	 Metadata	Required
Deploy Submit Application Flow	 Metadata	Required
Add 'Apply' Button on the Funding Program Layout	Other	Required
Install Outbound Funds NPSP Extension	 Managed Package	Optional
Create the Funding Program Portal	Other	Required
Publish the Funding Program Portal	Other	Required
Install Funding Program Portal Navigation Menu	 Metadata	Required
Add Grantseeker Profile to Funding Program Portal	Other	Required
Grant Visibility to Community Users	 Metadata	Required
Add Funding Program Portal Branding	 Metadata	Required





MetaDeploy and the Product Delivery Model

A product is much more than a package

A product can encompass a tremendous breadth of customization to create an immersive customer experience.

A product isn't just one package.

It's one, or four, or six, or 10 packages that work together or offer optional features.

It's setup automation that streamlines the first-login experience.

It's configuration that makes the product usable immediately.

MetaDeploy delivers products, not just packages.



CumulusCI and portable automation make sharing and collaboration easy – in one organization, across organizations, between roles, and in the open source community.





thank
you

BLAZE
YOUR
TRAIL

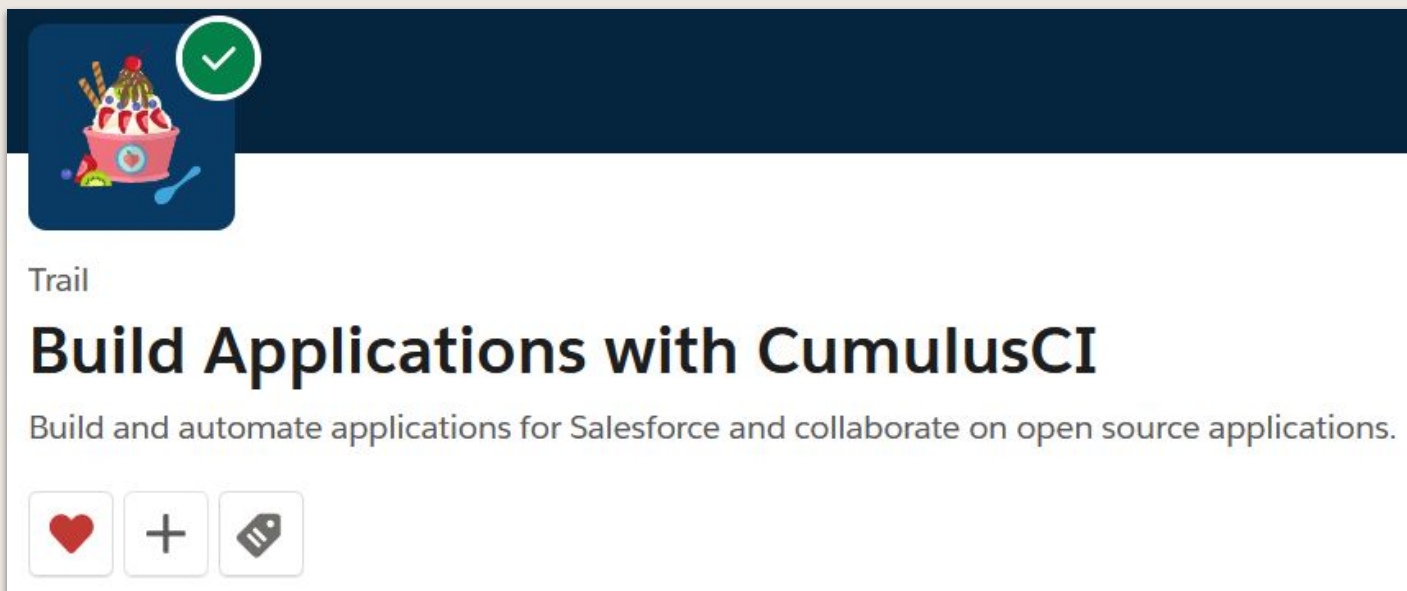
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Now available on Trailhead!



Build Applications with CumulusCI:

<https://trailhead.salesforce.com/en/content/learn/trails/build-applications-with-cumulusci>



The image shows a digital screen displaying a Trail card. The card has a dark blue header with a green checkmark icon in a white circle. Below the header is a white section with the word "Trail" in small text, followed by the title "Build Applications with CumulusCI" in large bold text. Under the title is a subtitle: "Build and automate applications for Salesforce and collaborate on open source applications." At the bottom of the card are three icons: a red heart, a plus sign, and a tag icon.

Trail

Build Applications with CumulusCI

Build and automate applications for Salesforce and collaborate on open source applications.

♥ + 🏷

Resources



CumulusCI Documentation: cumulusci.readthedocs.io

CumulusCI on the Trailblazer Community:

https://success.salesforce.com/_ui/core/chatter/groups/GroupProfilePage?g=0F93000000009M9Z

CumulusCI Open Source Project: github.com/SFDO-Tooling/CumulusCI

Salesforce.org Products and the Open Source Commons

- github.com/SalesforceFoundation/NPSP
- github.com/SalesforceFoundation/EDA
- github.com/SFDO-Community